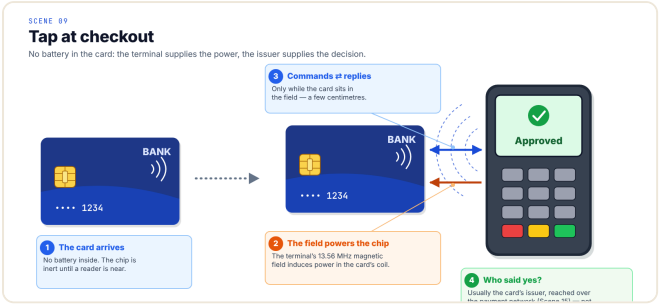
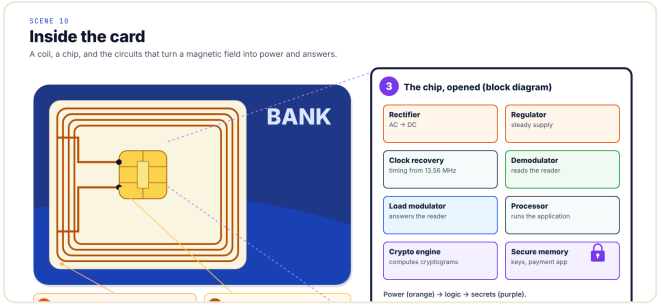
The Invisible Engine, in one page

A card with no battery, a tag that whispers, and the keys that decide who may read a location. Six pictures, in order. Each opens its scene in the full lesson.



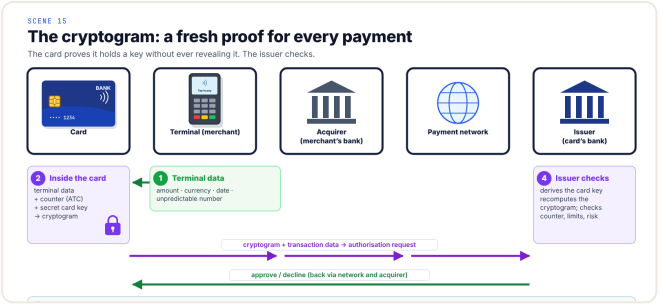
1 A tap with no battery

A child taps a card at a shop and asks how it works without a battery. The terminal's invisible magnetic field powers the card for a moment.



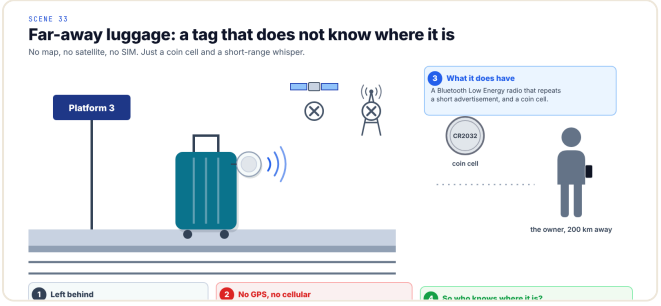
2 Open the card

Inside: a loop of fine wire around the edge and a tiny chip. The loop catches energy; the chip uses it to talk and keep a secret.



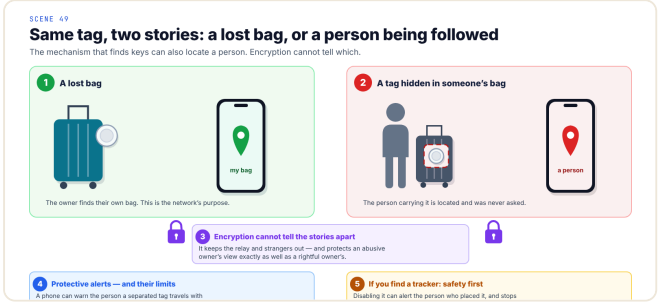
3 A fresh secret code

Each payment, the card makes a new code that only its bank can check. A copied code is useless next time.



4 A tracker that whispers

A lost suitcase holds a small tag. It cannot see a map. It whispers, and a stranger's phone that knows where it is passes the message on, sealed.



5 Keys, accounts, people

Whoever holds the right key can read where the tag has been. That protects the owner — and can expose a person carrying someone else's tag.



6 Waves, qubits, tomorrow's locks

Why is the code hard to break? The story reaches waves that cancel, qubits, and the new locks being built for a quantum future.

The big idea: energy, signals, data and promises about who may read the data are four different things. Most mistakes come from drawing them as one arrow.